

GABRIEL FRANCO

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Education

Boston University

Sep 2021 – Present

PhD in Computer Science

Boston, MA

- Advisor: Mark Crovella
- Research interests: Mechanistic Interpretability, Natural Language Processing (NLP), Causal Analysis, Concepts Representations and Geometry, LLM Evaluation, Learning from Label Proportions, Machine Learning/Deep Learning

Federal University of Viçosa

Aug 2018 – Jun 2021

MSc in Computer Science

Brazil

- Advisor: Giovanni Comarella
- Thesis: New Hyperparameter Strategies for Learning with Label Proportions

Federal University of Viçosa

Mar 2014 – July 2018

B.S. in Computer Science

Brazil

Publications

• Mechanistic Interpretability:

- * **Franco, Gabriel**, and Mark Crovella. “Pinpointing Attention-Causal Communication in Language Models”. **Accepted in NeurIPS 2025.**
- * **Franco, Gabriel**, and Mark Crovella. “Sparse Attention Decomposition Applied to Circuit Tracing”. <https://arxiv.org/abs/2410.00340>.

• LLM evaluation:

- * Calais, Pedro, **Gabriel Franco**, et al. “Disentangling Text and Math in Word Problems: Evidence for the Bidimensional Structure of Large Language Models’ Reasoning.” Findings of the Association for Computational Linguistics: ACL 2025. 2025.

• Learning from Label Proportions (LLP):

- * **Franco, Gabriel**, Giovanni Comarella, and Mark Crovella. “Evaluating LLP Methods: Challenges and Approaches”. <https://arxiv.org/pdf/2310.19065>
- * **Franco, Gabriel**, Mark Crovella, and Giovanni Comarella. “Dependence and Model Selection in LLP: The Problem of Variants.” Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. 2023.

• Others:

- * Ghaemi, Golsana, **Franco, Gabriel**, Kazem Taram, and Renato Mancuso. “Heterogeneous Memory Benchmarking Toolkit.” **Accepted in RTSS 25.**
- * **Franco, Gabriel**, Marcos Henrique Fonseca Ribeiro, and Giovanni Comarella. “Towards an interpretable metric for DOTA 2 players: An unsupervised learning approach.” 2019 8th Brazilian Conference on Intelligent Systems (BRACIS). IEEE, 2019.
- * Gomes, Andréia Patricia, et al. “Plasmodium Falciparum Infection: In Silico Preliminary Studies.” Abakós 5.1 (2016): 63-83.

Research & Industry Experience

Research Assistant

Sep 2021 – Present

Boston University

Boston, MA

- Towards understanding internal communication mechanisms in Large Language Models (LLMs): Developed a novel spectral decomposition method to identify causal communication circuits in LLMs, enabling new applications in circuit tracing and low-rank model interventions.
- Hyperparameter selection strategies and new datasets for Learning from Label Proportions (LLP): Proposing that LLP problems can be characterized into distinct variants, showing that different hyperparameter selection strategies are needed for each, and proposing a new set of benchmark datasets to account for these variants.

Data Scientist Intern

May 2024 – Aug 2024

Microsoft

Redmond, WA

- Successfully fine-tuned and optimized multi-modal SLMs for local image tagging, achieving 85% of a large foundation model’s performance while meeting < 5s inference targets.

Data Scientist

Sep 2020 – Jul 2021

SEEK

Brazil (Remote)

- Optimized a large-scale recommender system, reducing average response time by over 50% to support scalability for a growing user base.
- Designed and A/B tested a new job recommendation model, increasing click-through rate by 5% and job applications by 10%.

Data Scientist

Jul 2020 – Sep 2020

Localiza

Brazil (Remote)

- Developed and validated a machine learning model to identify reliable customers; presented feasibility analysis and results to stakeholders to inform business decision-making.

Master's Student Researcher

Aug 2018 – Jul 2020

CAPES (Brazilian Government Research Funding Agency)

Brazil

- New Hyperparameter Strategies for Learning with Label Proportions (LLP): Worked on improving the cross-validation strategies for LLP, which is a problem that the instances do not have individual labels.

Undergraduate Student Researcher

Mar 2015 – Jul 2018

CNPq/FAPEMIG (Brazilian Government Research Funding Agencies)

Brazil

- Developed the Bio-Oracle software: Bio-ORACLE is a software written in JAVA which uses data mining techniques to help decision-making in bioethics. Bio-ORACLE was developed at the Laboratory of Epidemiological and Computational Methods in Health of Department of Medicine and Nursing in Federal University of Viçosa.
- Implementing malaria and plasmodium on Autosimmune: Autosimmune is a multiagent human immune system simulator made in JAVA. Autosimmune was developed at the Laboratory of Epidemiological and Computational Methods in Health of Department of Medicine and Nursing in Federal University of Viçosa.

Other Academic Productions

- Siqueira-Batista, Rodrigo, et al. Parasitologia: Fundamentos e Prática Clínica. Guanabara, 2020. ISBN 9788527735735. (I co-wrote Chapter 4 about the computational approach in the study of parasitic diseases)
- Comarela, G.; Franco, G. ; Trois, C. ; Liberato, A. ; Martinello, M. ; Corrêa, J. H. ; Villaça, R. Introdução à Ciência de Dados: Uma Visão Pragmática utilizando Python, Aplicações e Oportunidades em Redes de Computadores SBRC 2019 (Short Course)

Technical Skills

Languages: Python, C/C++, Java.

ML/DS Libraries: PyTorch, Transformers, PEFT, Scikit-learn, Pandas, Numpy, Matplotlib, Seaborn, TransformerLens, nnsght.

Other Tools: Git, Linux, SQL, Redis, ElasticSearch.

Academic Service

- Reviewer ICLR 2026
- Reviewer Mechanistic Interpretability Workshop at NeurIPS 2025
- Reviewer NeurIPS 2025
- Reviewer ICLR 2025

Awards & Grants

- NeurIPS 2025 Scholar Award
- KDD'23 Student Travel Award
- KDD'22 Student Travel Award
- CNPq Research Scholarship (PIBIC/CNPq) - 2018
- CNPq Research Scholarship (PIBIC/CNPq) - 2017
- CNPq Research Scholarship (PIBIC/CNPq) - 2015
- FAPEMIG Research Scholarship (PIBIC/FAPEMIG) - 2015

Teaching Experience

Teaching Assistant

Boston University

- CAS CS 132: Geometric Algorithms.

Jan 2025 – May 2025

Boston, MA

Teaching Assistant

Boston University

- CDS DS 701: Tools for Data Science.

Sep 2023 – Dec 2023

Boston, MA

Instructor

Federal University of Viçosa

- L^AT_EX short course.

Nov 2019

Brazil

Teaching Assistant

Federal University of Viçosa

- INF 100 - Introduction to Programming I.

Mar 2019 – Jul 2019

Brazil

Relevant Coursework

- CS542 - Machine Learning
- CS565 - Algorithmic Data Mining
- DS563 - Algorithmic Techniques for Taming Big Data
- CS505 - Introduction to Natural Language Processing
- LX690 - Metrics and Evaluation in Natural Language Processing (Audited)

Languages

- Portuguese (Native or bilingual proficiency)
- English (Professional working proficiency)
- Spanish (Elementary proficiency)

Additional Activities

- ACM International Collegiate Programming Contest - Regionals - 2014, 2015, 2017
- Minas Gerais State Programming Contest - 2014, 2015, 2017

Referees

- Prof. Mark Crovella
Boston University
crovella@bu.edu
- Prof. Evimaria Terzi
Boston University
evimaria@bu.edu
- Prof. Giovanni Ventorim Comarela
Universidade Federal do Espírito Santo
gc@inf.ufes.br